Engineering Chemistry (AHT-002) - Unit 1

Unit 1 - Atomic & Molecular Structure - notes in easy English, matched to the UTU syllabus.

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What this unit covers (syllabus): atomic theory and nuclear models, Heisenberg's uncertainty principle, de Broglie waves, the Schrödinger equation for hydrogen, VBT and MOT of diatomic molecules (H_2, O_2, N_2, H_2O, NH_3, XeO_3, XeO_4), crystal field theory and magnetic properties of transition metal ions, and band theory of metals.

Key ideas: de Broglie: every moving particle has a wave, $\lambda = h/mv$. **Heisenberg:** $\Delta x \cdot \Delta p \geq h/4\pi$ - position and momentum cannot both be exact. The **Schrödinger equation** ($H\psi = E\psi$) gives orbitals as solutions. **VBT**: bonds form by orbital overlap, explained with hybridisation (sp^3 in NH_3/H_2O). **MOT:** atomic orbitals combine into bonding (lower energy) and antibonding (higher) molecular orbitals; **bond order** = (bonding - antibonding electrons)/2 - O_2 has bond order 2 and is paramagnetic (two unpaired electrons), the classic question. **Crystal field theory:** d-orbitals split in a ligand field (t_2g/e_g in octahedral); strong-field ligands pair electrons (low spin), weak-field leave them unpaired (high spin) - this decides magnetism. **Band theory:** metals conduct because their valence and conduction bands overlap.

Exam pointers: MO diagram of O_2 or N_2 with bond order and magnetism is the most repeated question. de Broglie/Heisenberg numericals (plug into the formula) and "VBT vs MOT" are regular theory.

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